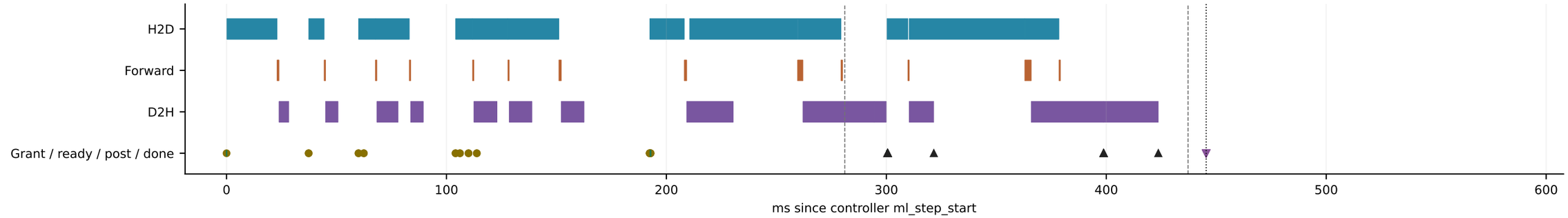
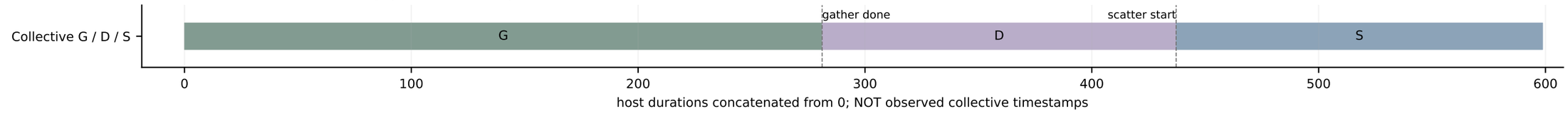
mlp | 24c1g | numa | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

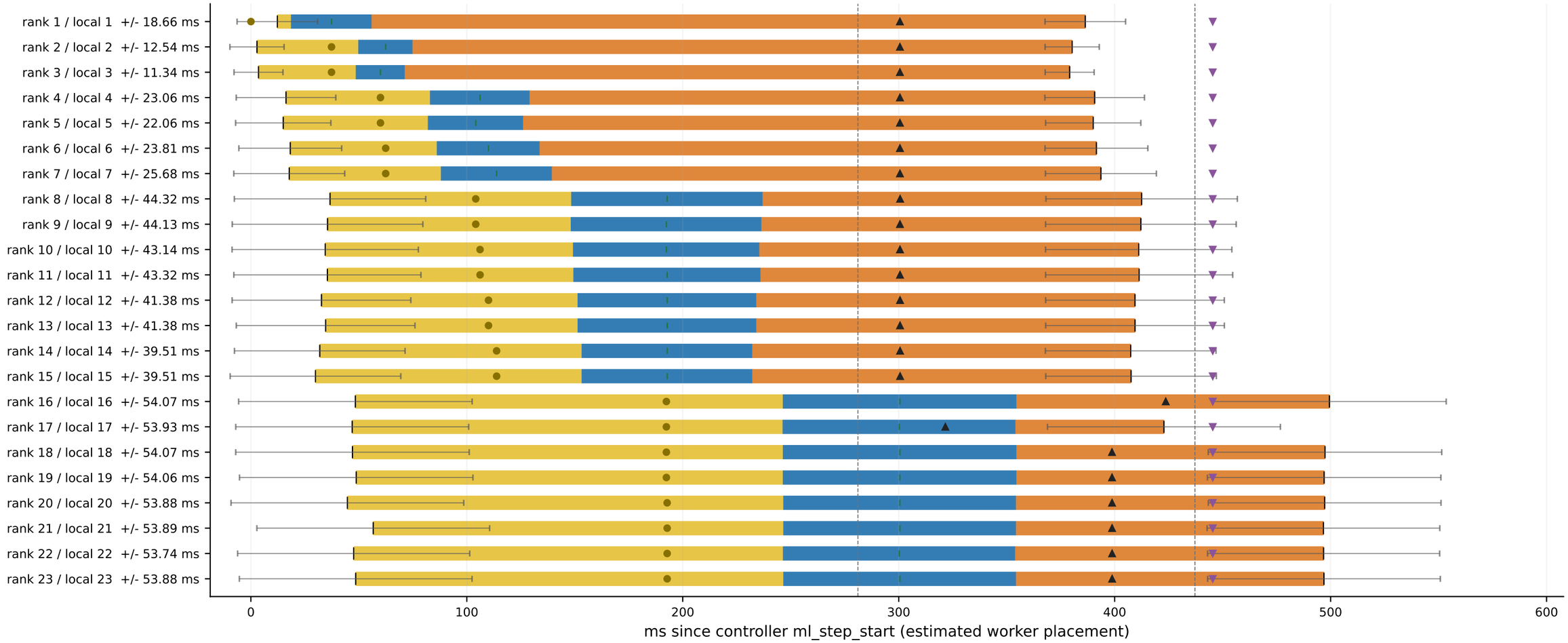
Controller rank 0 (WG-local 0): own observed host/GPU lanes



Schematic collective reference (gaps omitted, independent zero)



23 workers: causal midpoint placement; whiskers = offset bounds



Credit wait + logging gap
Blocking send + ACK
Result wait + receive
Grant logged
Ready (before ACK)
Result send post
Done (after Waitall)
Own ML start/end
Schematic G end / S start